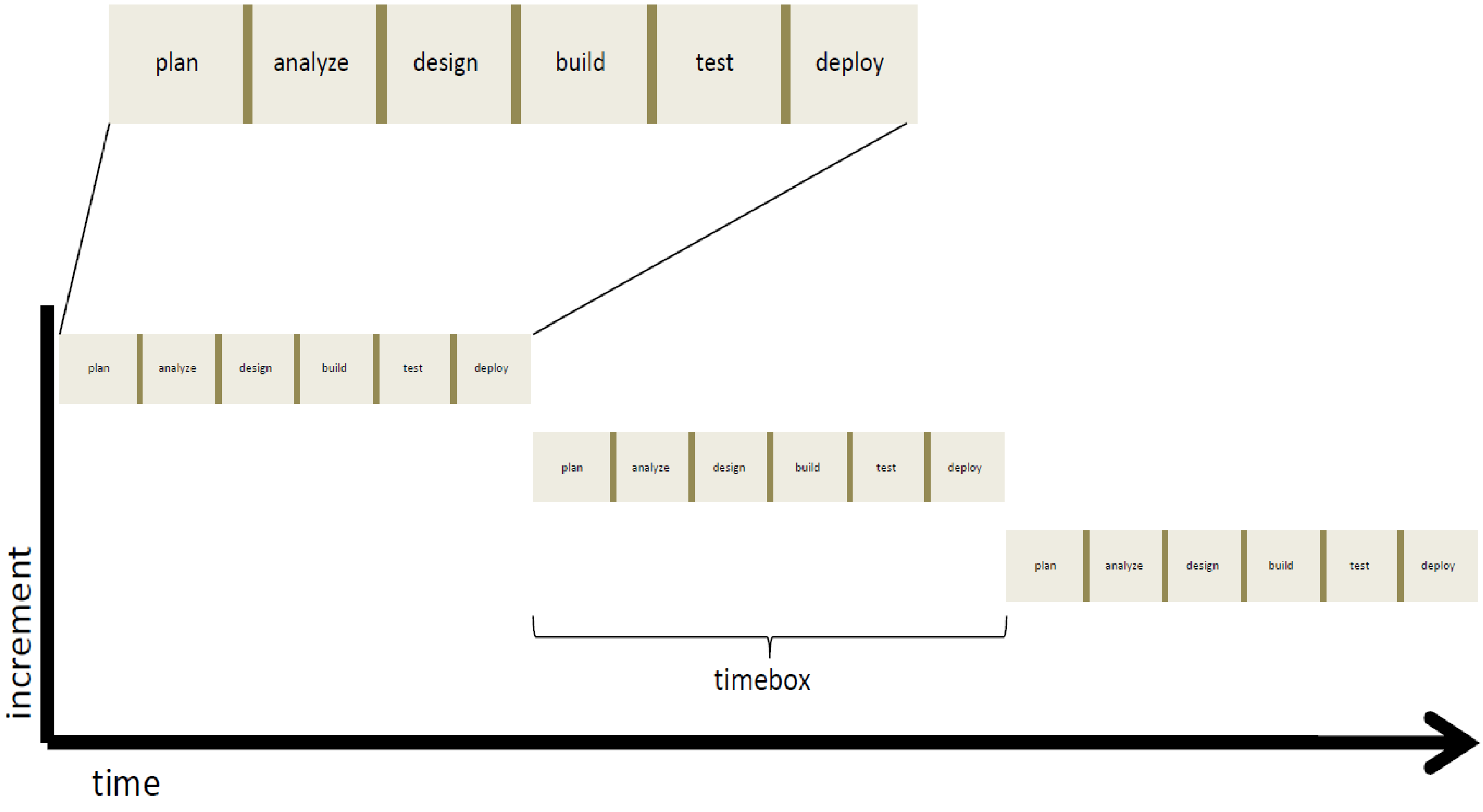
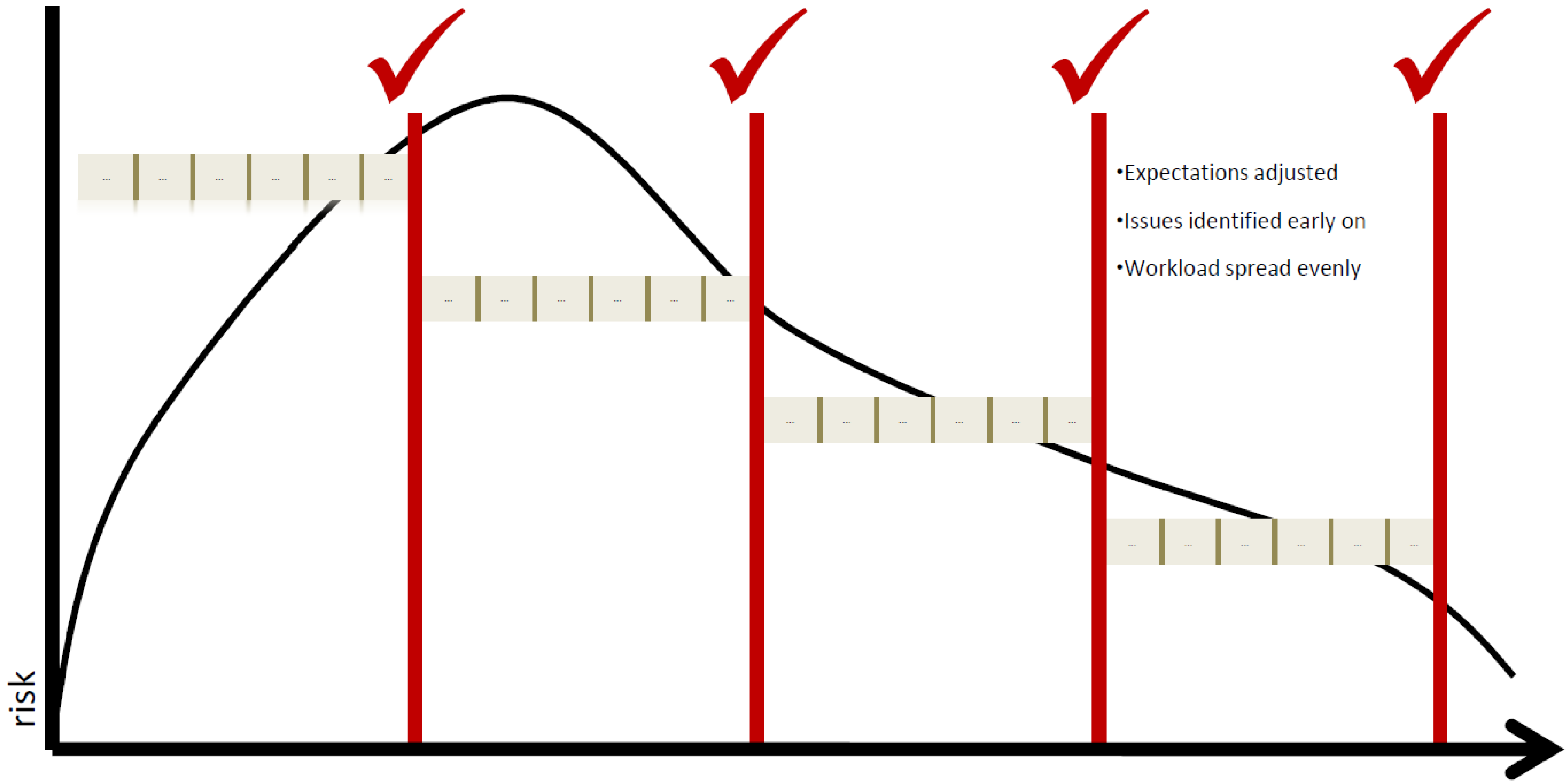
Agile Development Model

- An evolutionary, value-driven and time boxed development model
- Inspired by evolutionary and adaptive behavior
- Consists of several iterations of development cycles each contributing to an incremental growth of new capabilities
- An iteration consists of an entire cycle of plan, analyze, design, build, test, and deploy activities
- An iteration results in production-ready capabilities
- Iterations are time-boxed to a fixed duration

Agile Development Model



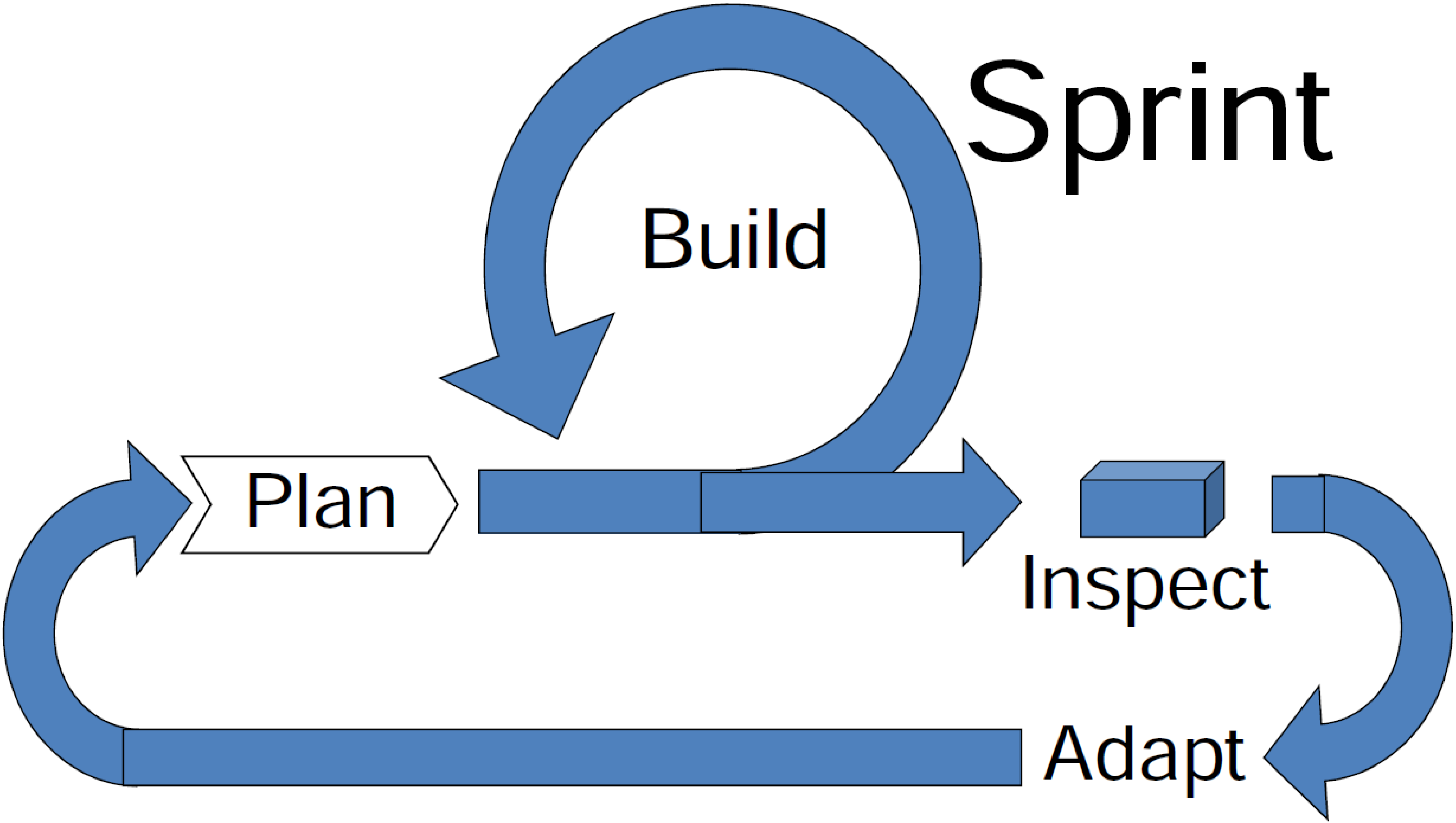
Agile Development Model



Agile Manifesto

Individuals and interactions	Over	Process and tools
Working software	Over	Comprehensive documentation
Customer collaboration	Over	Contract negotiation
Responding to change	Over	Following a plan

Agile Development Model



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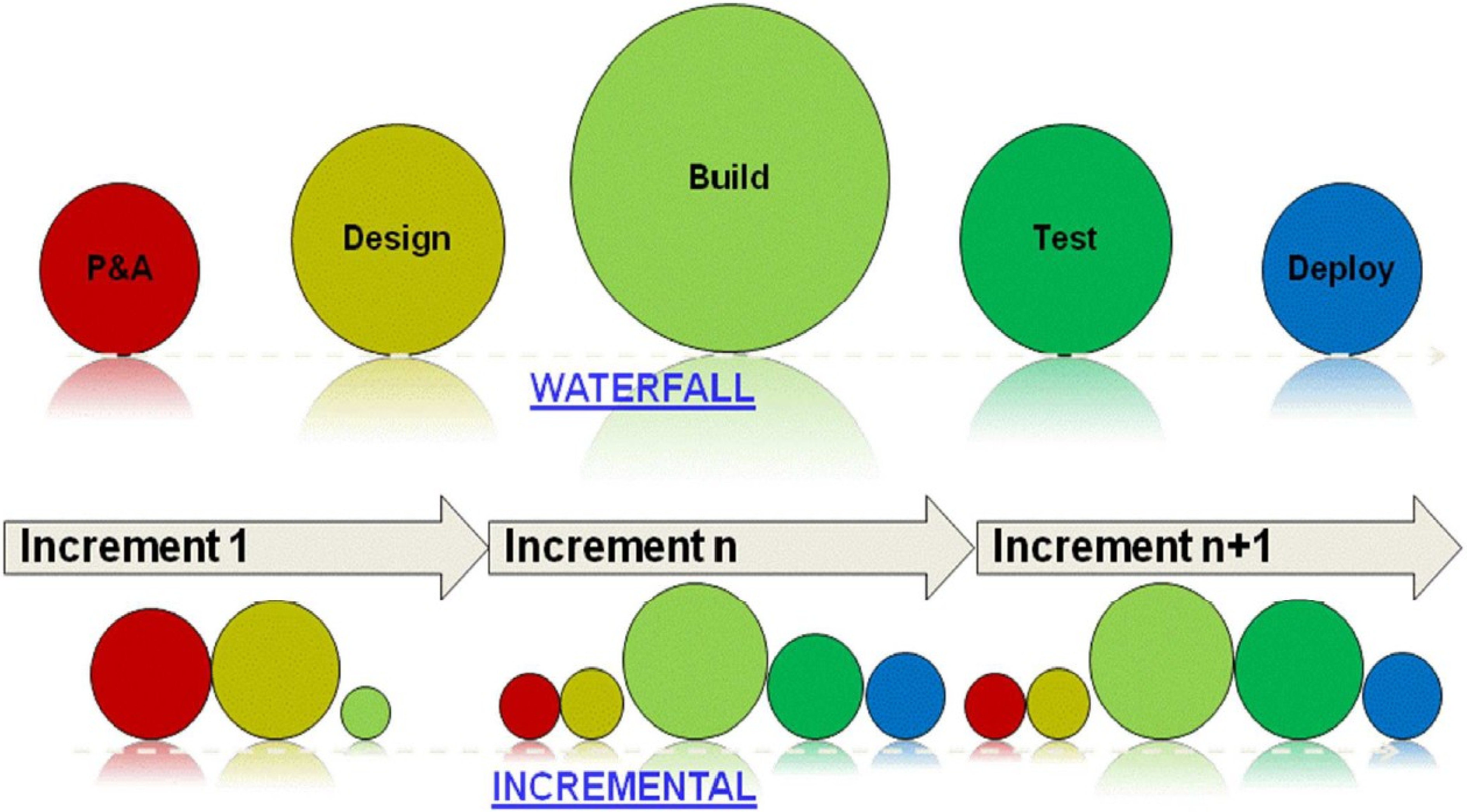

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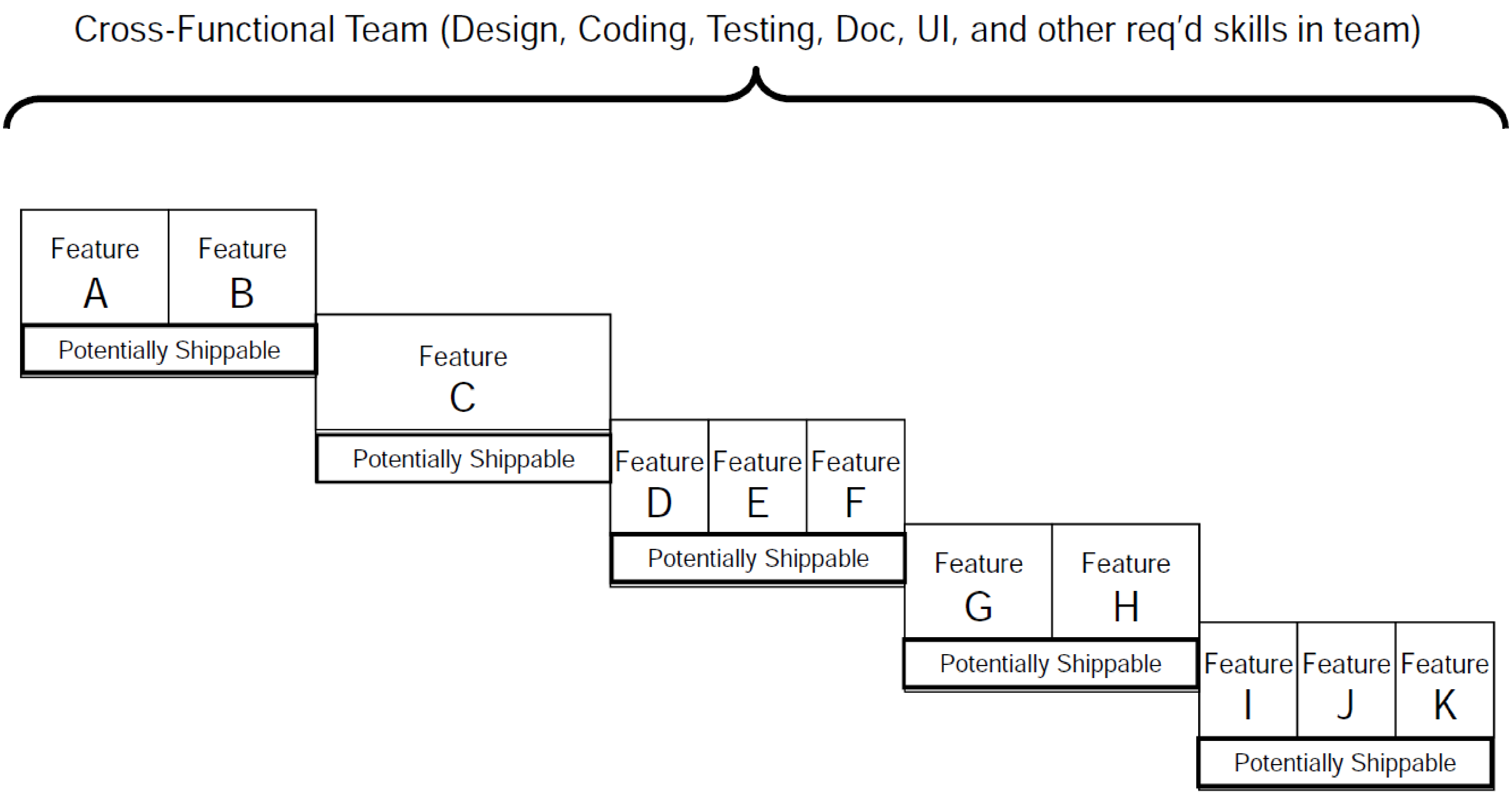
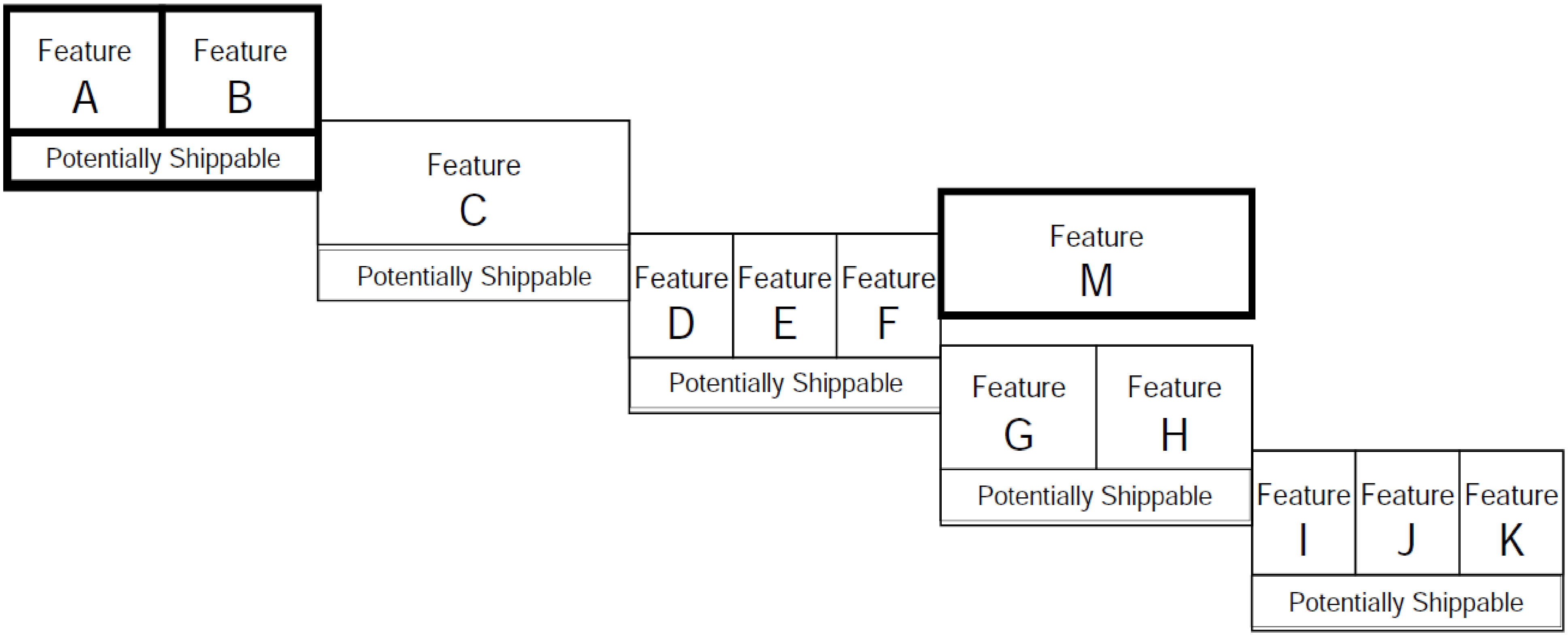


Agile Development Model

- We can accommodate change during the development process.



Prioritizing work and Tracking Progress

- Requirements should be prioritized with the customer based on their value and risk profile.
- Project Progress should be measured.

Scrum

- Scrum roles
- Scrum Process
- Requirements
- Prioritising and Measuring Work

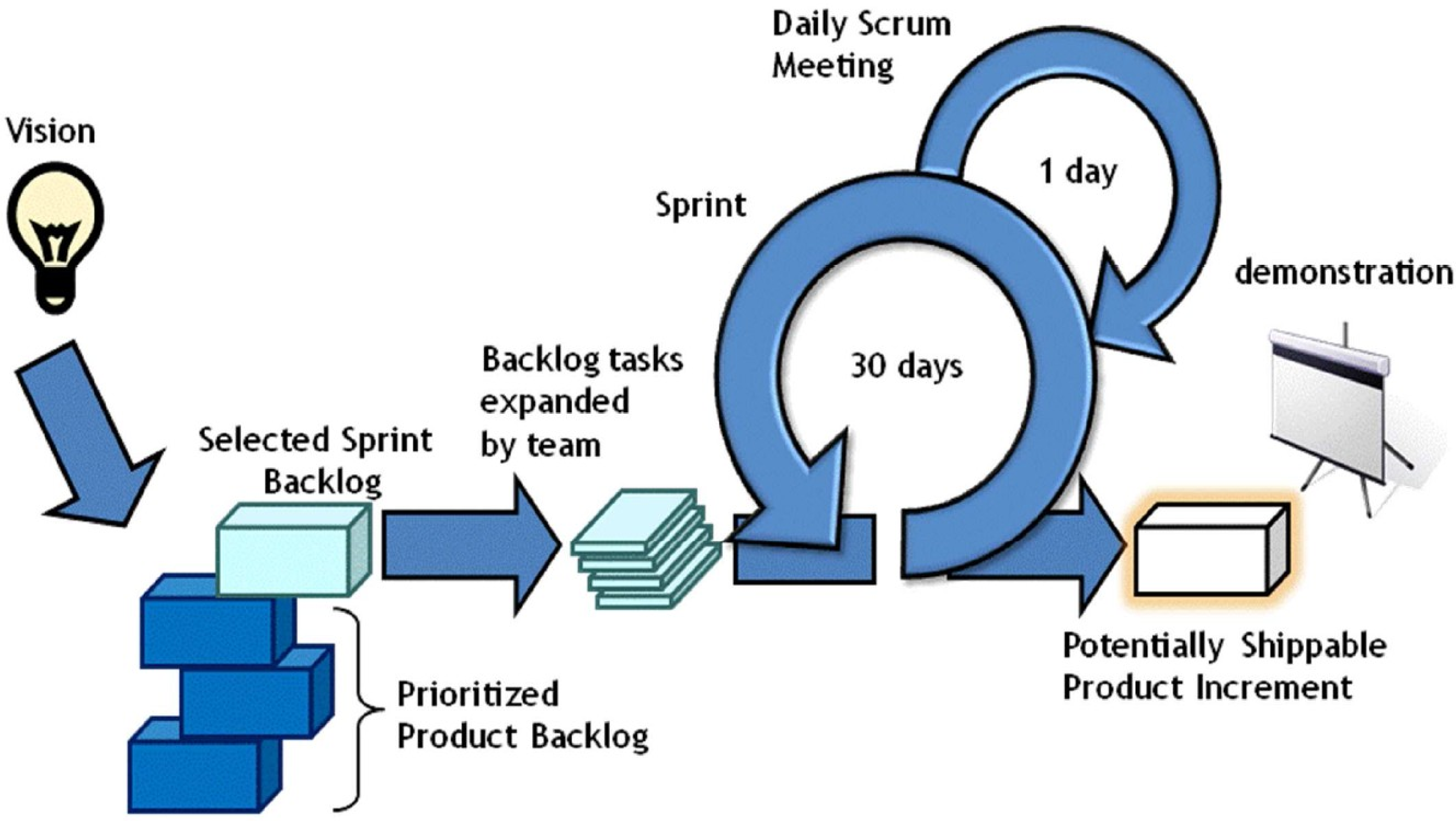
Scrum: An Overview

- SCRUM is an empirical process that deals with high complexity development
- Promotes empirical processes instead of defined processes
- Scrum is a simple but powerful framework for teams and customers to inspect and adapt as product is produced
- Promotes a high degree of clarity and transparency to everyone involved
- Scrum rapidly surfaces dysfunction, and enables teams and organizations to continuously improve their effectiveness
- Scrum exposes issues but does not provide the answers for fixing issues
- The Scrum framework can be used with technology specific implementation methodology

The Daily Scrum

- Allows the team to self-organize & synchronize on tasks
- Maximum 15 minutes
- Three questions to answer:
  1. What have you done yesterday?
  2. What are you working on today?
  3. What impediments are getting in your way?
- Things to avoid:
  - Telling the team what to do
  - Creating an atmosphere of fear
  - Have outsiders interfere
  - Providing direction without team involvement

Scrum



Daily Scrum Meeting
Vision
1 day
Sprint
demonstration
Backlog tasks expanded by team
30 days
Selected Sprint Backlog
Potentially Shippable Product Increment
Prioritized Product Backlog

The First Increment

- May run a bit longer than subsequent increments to allow for sufficient preparation time but should be time boxed.
- Key activities include planning and analysis tasks as well as some key design tasks
- At the end of the first increment, requirements should be sufficiently detailed to support planning and scope management.
- Detailed requirements definition can be scheduled in increments.

Increment Length

- Shorter increments provide more frequent feedback but may increase overhead
- Most agile methodologies recommend an increment length between two weeks and one month.
- SCRUM recommends a 30 day increment length.
- Considerations for selecting an increment length:
  - Complexity of the solution
    - Highly complicated solutions may require more time to coordinate deployments.
    - Deployment automation is critical.
  - Maturity of the team
    - A team may need time to get used to fast paced development.
    - Shorter increments may increase pressure which can lead to a “throw it over the wall” mentality
  - Overall project timeline
    - If the development time is less than 6 months, it’s usually better to go with 2 week increments.
- Once you select an increment length, you should stick with it to establish a routine.

Scrum Roles

- Team
  - 7 +/- 2 People
  - Cross-functional
  - Self-organizing
  - 2 key freedoms, where possible
    - What to work on in a Sprint
      - Something familiar, something new...
    - How much time needed for a task
      - Line managers no longer imposing deadlines

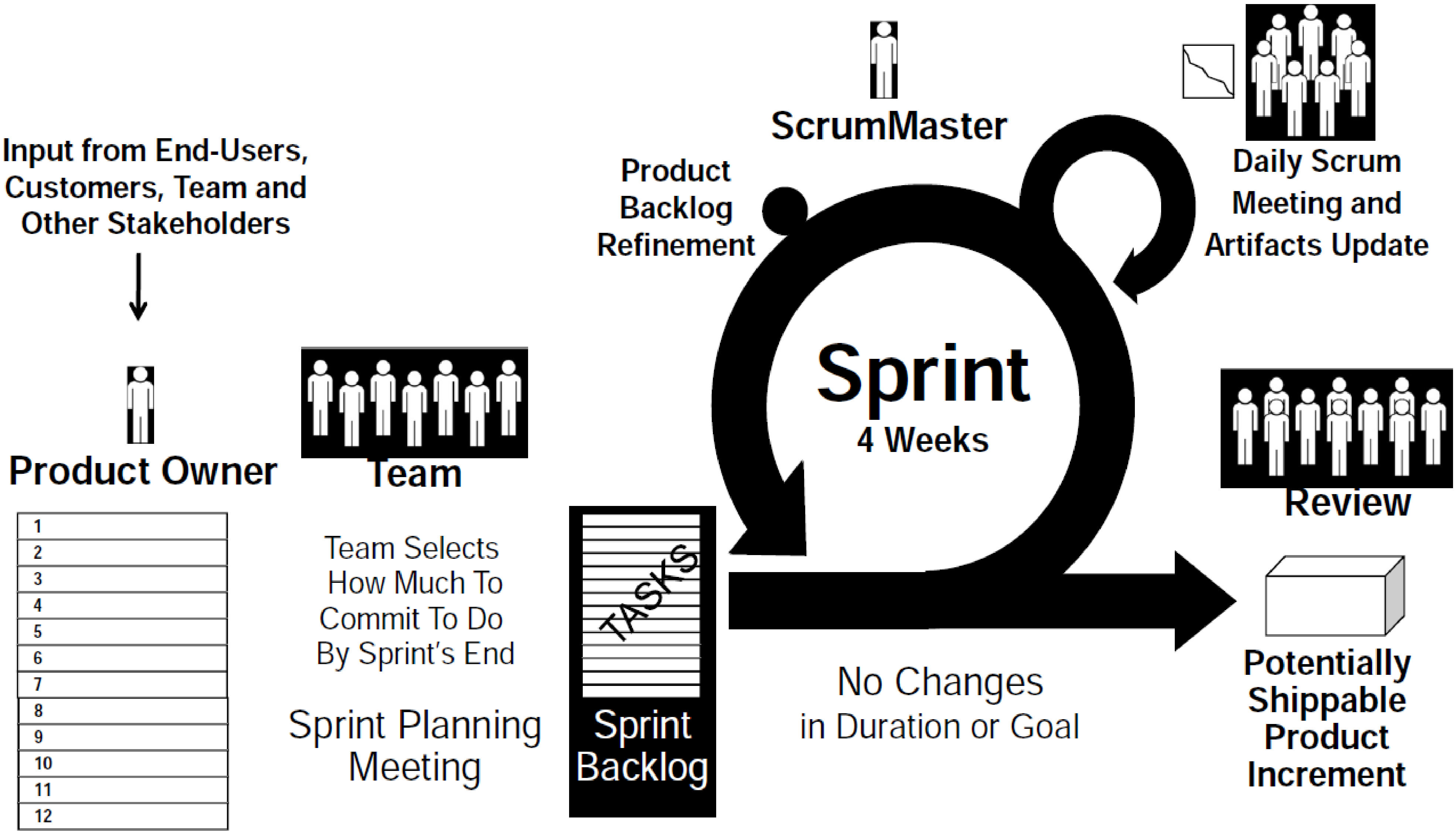
Scrum Roles

- Scrum Master
  - Responsible ensuring the Scrum process is understood and followed
    - 3 P's:
      - Process Owner
        - Teaches Scrum to Team, PO, and stakeholders
        - Coaches the Team, PO, and stakeholders to achieve maximum value and ROI using Scrum
      - Problem Solver
        - Helps remove Team's constraints and impediments ("blocks")
      - Protector
        - Protects the Team from disruption or interference

Scrum Roles

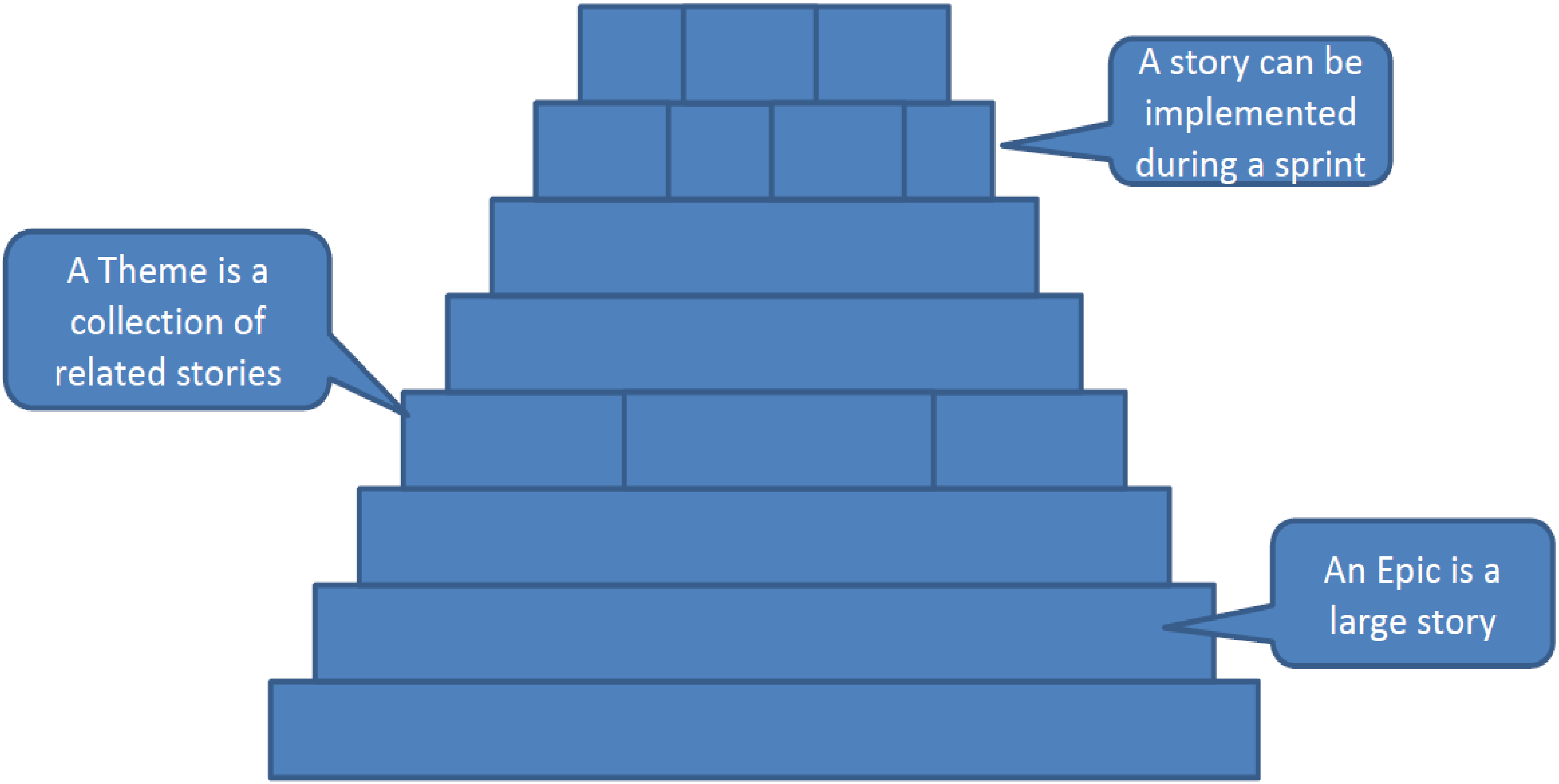
- Product Owner
  - Responsible for maximizing the value of the work that the Team does
  - Owns the vision and overall goals
  - Owns the prioritized list of what needs to be produced to achieve maximum value and ROI (the Product Backlog)
  - Decides when product is ready to ship

Scrum Process



Product Backlog

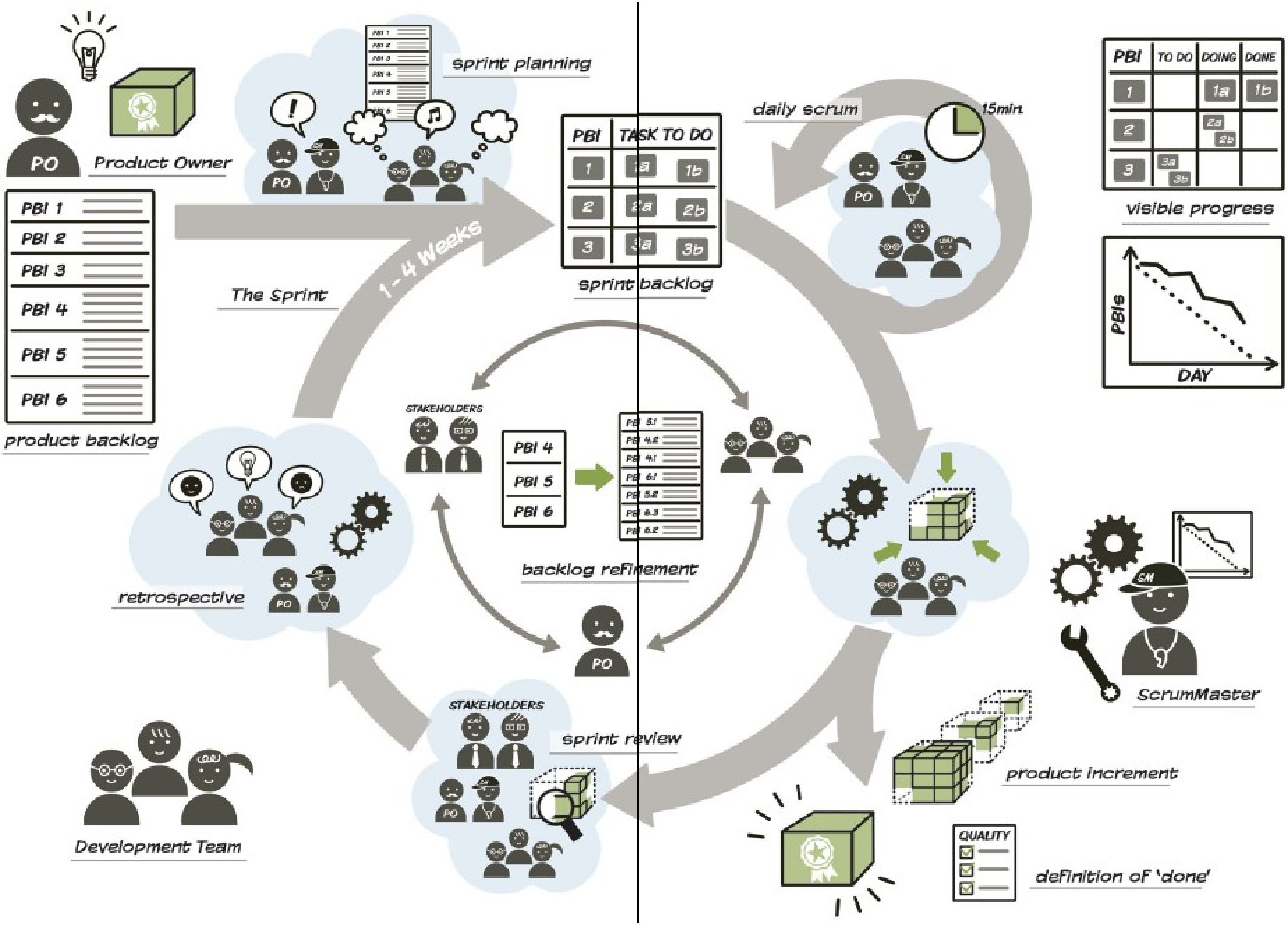
- Product backlog is made up of stories



Summary

- Focus on software deliverables
  - Can be reviewed and tested
  - Help improve quality of estimates
- Understanding Scrum
  - Scrum Roles, Scrum Process, Requirements
  - Prioritising work that brings value and Measuring Work progress
- What are User Stories?
  - Users and user roles
  - Exercises
  - Why User Stories?

Scrum Flow



Product Owner
PO
product backlog
PBI 1
PBI 2
PBI 3
PBI 4
PBI 5
PBI 6
sprint planning
sprint backlog
PBI
TASK TO DO
1
1a
1b
2
2a
2b
3
3a
3b
daily scrum
15min.
TO DO
DOING
DONE
visible progress
PBIs
DAY
The Sprint
1 – 4 Weeks
STAKEHOLDERS
backlog refinement
retrospective
sprint review
product increment
ScrumMaster
SM
Development Team
QUALITY
definition of 'done'